THE BIT DELIVERED TWICE: TWO KINDS AND NO THIRD

STEP 10 · THE BIT DELIVERED TWICE, BY ITS TWO KINDS, AND NO THIRD

KIND 1 · THE FORMAL CARGO

The record's value carried to the arithmetic frame. It arrives free, the antecedent a theorem; it gives the Goldbach statement in one step; and it is exactly the statement.

`FormalCargo` · `cargo_arrives` · `formal_kind_delivers`
`formal_iff_statement`

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KIND 2 · THE PHYSICAL WITNESS

Every examination of an even number is a physical run of the search, registered at the Landauer floor; a run ending with no split would leave an unsplit record, and the carrier registers none. Its last field is the bit read in the register of heat, carried as a field, never proved.

`PhysicalWitness` · `witness_closes` · `physical_kind_delivers`
`physical_iff_statement`

The two kinds are one bit
The physical witness is a formal cargo, and the cargo is exactly the statement.
`the_two_kinds_are_one_bit`

Two kinds and no third
Every witness of the statement, in any vocabulary, is a delivery of both kinds.
`two_kinds_and_no_third`
`every_route_is_both_kinds`

The one refuter
One even number at least four with no split refutes both kinds at once, and the statement itself.
`the_one_refuter` · `refuter_denies_the_sentence`

THE WITNESSES OF THE SUBJECT, TYPED · EACH STRICTLY INSIDE THE BIT OR SHORT OF IT

The certified region to $4 \cdot 10^{18}$
Carried as a field: load-bearing below its height, never reaching above it.
`certificate_is_load_bearing_arith`
`certificate_does_not_reach_above_arith`

The ternary and Chen theorems
Their shared shape does not give the binary: on an abstract frame both hold and the binary fails.
`weaker_do_not_give_binary`

The binary gives the ternary
In Mathlib's vocabulary the binary statement gives the ternary; the shape the ternary shares never gives the binary.
`binary_gives_ternary`

READING theorem on no axiom at all · binding theorem on propext (with Quot.sound) · binding theorem on Lean's standard axioms · definition or structure